

Characterizing the Noise-Induced Divergence Depth in QAOA for MaxCut: A Hardware-Simulation Benchmarking Study

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Abstract

The Quantum Approximate Optimization Algorithm (QAOA) is among the most studied candidates for practical utility on near-term, noisy quantum hardware. Yet a fundamental question remains open: at what circuit depth does hardware noise overwhelm the performance gains of additional QAOA layers? We introduce the concept of *divergence depth* p^* (the smallest circuit depth p at which hardware approximation ratio diverges from noiseless simulation by more than a threshold $\delta = 0.05$) and present the first systematic empirical measurement of p^* as a function of problem size n . We benchmark QAOA applied to the MaxCut problem on 3-regular and Erdős-Rényi random graphs with $n \in \{6, 8, 10, 12\}$ qubits, sweeping circuit depth p from 1 to 5. We evaluate two conditions: exact noiseless statevector simulation and a calibrated depolarizing noise model representative of IBM Eagle R3 superconducting hardware ($\varepsilon_{2Q} = 0.5\%$, $\varepsilon_{1Q} = 0.1\%$, $\varepsilon_{RO} = 1.0\%$). Across 30 random instances per (n, p) pair, we compute approximation ratios with 95% confidence intervals and compare against

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classical baselines including the Goemans-Williamson SDP ($\text{AR} \geq 0.8786$) and greedy assignment. The noisy approximation ratio peaks at $p^* = 4$ across all tested system sizes and declines thereafter, consistent with the $\mathcal{O}(\varepsilon np)$ noise accumulation model. The divergence depth framework gives practitioners a concrete tool for calibrating circuit depth budgets on near-term devices.

1 Introduction

The past decade has seen rapid growth in quantum hardware capabilities, with superconducting processors now routinely achieving gate fidelities above 99% for single-qubit operations and 99.5% for two-qubit gates on leading platforms [1]. Yet we remain firmly in what Preskill termed the Noisy Intermediate-Scale Quantum (NISQ) era [1]: devices of 50–1000 qubits operating without error correction, where coherent gate errors, decoherence, and measurement noise accumulate with circuit depth. The central challenge is identifying which algorithms can extract practical value from current hardware before noise overwhelms the computation.

The Quantum Approximate Optimization Algorithm (QAOA), introduced by Farhi, Goldstone, and Gutmann in 2014 [2], has emerged as a leading candidate. QAOA is a hybrid quantum-classical variational algorithm that applies alternating layers of a problem-dependent cost unitary and a mixing unitary, parameterized by angles γ and β . At circuit depth $p = 1$ applied to MaxCut on 3-regular graphs, QAOA achieves a provable approximation ratio of at least 0.6924 [2], and simulations show this approaches 1 as $p \rightarrow \infty$ [3]. These theoretical properties make QAOA appealing for combinatorial optimization on near-term devices.

However, a gap persists between theoretical performance and hardware reality. Gate errors accumulate with each QAOA layer: theory predicts the additive performance loss scales as $\mathcal{O}(\varepsilon np)$, where ε is the gate error rate, n is the number of qubits, and p is the circuit depth [4]. Empirical benchmarks confirm degradation: Harrigan et al. [5] found hardware approximation ratio plateaued at $p = 2$ –3 on Google Sycamore, while Lotshaw et al. [6] observed performance at $p = 3$ actually falling *below* $p = 1$ on IBM hardware for large instances. Despite these scattered observations, no prior work has operationalized the *divergence depth* p^* (the maximum QAOA circuit depth before noise-induced degradation becomes dominant) as a primary, systematically measured quantity.

We make two contributions. First, we introduce and empirically measure $p^*(n)$ across a grid of problem sizes and circuit depths using a consistent methodology: 30 random instances per (n, p) pair, exact optimal cut computation for ground-truth approximation ratios, and 95% confidence intervals. Second, we test whether IBM’s **FakeSherbrooke** calibration-based noise model reliably predicts hardware performance at this scale.

The remainder of this paper is organized as follows. Section 2 provides background on MaxCut, QAOA, classical baselines, and prior benchmarking work. Section 3 describes our experimental methodology in full reproducible detail. Section 4 presents our results. Section 5 discusses the physical interpretation of our findings and their implications for practitioners. Section 6 concludes.

2 Background and Related Work

2.1 The MaxCut Problem

Given an undirected graph $G = (V, E)$ with $|V| = n$ vertices and $|E| = m$ edges, the MaxCut problem asks for a partition $(S, \bar{S})$ of V that maximizes the number of edges with one endpoint in each part:

$$\text{MaxCut}(G) = \max_{x \in \{0,1\}^n} \sum_{(u,v) \in E} \frac{1 - (-1)^{x_u + x_v}}{2}.$$

MaxCut is NP-hard in general [7]. It admits the following classical approximation hierarchy:

- **Random assignment:** AR = 0.5 in expectation, since each edge is cut independently with probability 1/2.
- **Greedy algorithms:** Iteratively assigning each vertex to maximize the current cut achieves AR ≈ 0.55 –0.6 on random graphs.
- **Goemans-Williamson (GW) SDP [7]:** Semidefinite programming with random hyperplane rounding achieves AR ≥ 0.8786 in the worst case, with AR ≈ 0.93 –0.95 typical on random graphs. Under the Unique Games Conjecture, this ratio is optimal for polynomial-time classical algorithms.

Barak et al. [8] proved that classical local algorithms can achieve AR ≥ 0.725 on high-girth 3-regular graphs, *exceeding* the QAOA $p = 1$ guarantee of 0.6924 [2] on those instances. We do not claim quantum advantage over classical baselines; our framing is diagnostic, not competitive.

2.2 The Quantum Approximate Optimization Algorithm

QAOA [2] prepares the state

$$|\gamma, \beta\rangle = U(B, \beta_p)U(C, \gamma_p) \cdots U(B, \beta_1)U(C, \gamma_1)|+\rangle^{\otimes n},$$

where $U(C, \gamma) = e^{-i\gamma C}$ is the cost unitary for the MaxCut Hamiltonian $C = \sum_{(u,v) \in E} (I - Z_u Z_v)/2$, and $U(B, \beta) = e^{-i\beta \sum_i X_i}$ is the mixer unitary. The angles $(\gamma, \beta) \in \mathbb{R}^{2p}$ are optimized classically to maximize the expected cut value $\langle \gamma, \beta | C | \gamma, \beta \rangle$.

In practice, the cost unitary is implemented as $RZZ(2\gamma_k)$ gates per edge and the mixer as $RX(2\beta_k)$ gates per qubit. For 3-regular MaxCut at $p = 1$, the analytically optimal approximation ratio is $\text{AR} \geq 0.6924$ [2]. Zhou et al. [3] demonstrated via simulation that $p = 11$ achieves $\text{AR} \approx 0.94+$ on random 3-regular graphs, and introduced the TQA (Trotterized Quantum Annealing) initialization strategy: $\gamma_k = (k/p)\gamma_{\max}$, $\beta_k = (1 - k/p)\beta_{\max}$, which significantly outperforms random initialization at moderate p .

2.3 Noise and Hardware Benchmarking

Hardware execution introduces several error sources absent from ideal simulation. Two-qubit (entangling) gate errors dominate on current superconducting devices: IBM Eagle R3 processors achieve median ECR gate error rates of ~ 0.3 – 0.5% per gate. Readout errors contribute ~ 1 – 2% per qubit. Thermal relaxation (T_1/T_2 decoherence) is subdominant for short circuits but grows with depth. Stilck França and Garcia-Patron [4] proved rigorously that for circuits with T two-qubit gates at error rate ε , the achievable optimization accuracy degrades by $\Omega(\varepsilon T)$. For QAOA on n -qubit, m -edge graphs at depth p , $T = mp$ gates, so error grows as $\mathcal{O}(\varepsilon mp)$.

Wang et al. [9] showed that noise induces barren plateaus: gradient variance scales as $\mathcal{O}(e^{-\alpha np})$ under depolarizing noise, making optimization exponentially harder with circuit size. This further motivates the existence of a practical depth ceiling p^* beyond which neither optimization nor circuit fidelity are reliable.

Prior empirical benchmarks provide scattered evidence of this ceiling. Harrigan et al. [5] ran QAOA on Google Sycamore ($n = 23$, $p \leq 3$) and found hardware $\text{AR} \approx 0.735$ versus ideal $\text{AR} \approx 0.834$ at $p = 3$, with essentially no improvement from $p = 2$ to $p = 3$. Lotshaw et al. [6] found that on IBM hardware ($n = 16$, $p = 3$), AR fell to ≈ 0.55 – 0.62 versus ideal 0.69 . Weidenfeller et al. [10] showed AR decreases monotonically with n at fixed p on IBM Montreal. Pelofske et al. [11] provided the most comprehensive cross-platform comparison to date, finding substantial differences between IBM, IonQ, and Quantinuum hardware. Shaydulin et al. [12] claimed scaling advantage on IonQ Forte ($n = 32$) at $p = 1$.

Despite this body of work, no paper has defined or fitted $p^*(n)$ as a primary output, nor systematically validated the FakeSherbrooke noise model as a predictive tool. For comprehensive reviews, see Bharti et al. [13] and Blekos et al. [14].

3 Methodology

3.1 Graph Instance Generation

We study two graph families. Our primary benchmark uses **3-regular (cubic) graphs**, which have a fixed degree of 3 per vertex. This ensures consistent problem structure across different n values and enables comparison to the analytical $p = 1$ bound of 0.6924. Instances are generated using `networkx.random_regular_graph(3, n, seed=i)` for $i \in \{0, \dots, 29\}$ and $n \in \{6, 8, 10, 12\}$.

As a secondary analysis, we include **Erdős-Rényi random graphs** $G(n, 0.5)$, which provide variable-degree instances and test the generalizability of our findings beyond the structured 3-regular family.

We use exactly 30 seeded instances per $(n, \text{graph type})$ combination. For each instance, the exact maximum cut is computed by brute-force enumeration of all 2^n bitstrings, which is trivial for $n \leq 12$ (at most $2^{12} = 4096$ evaluations), so approximation ratios $\text{AR} = \langle C \rangle / \text{MaxCut}(G)$ are exact rather than estimated.

3.2 QAOA Circuit Construction and Parameter Optimization

We implement the standard QAOA ansatz using $RZZ(2\gamma_k)$ gates per edge for the cost unitary and $RX(2\beta_k)$ gates per qubit for the mixer, for $k = 1, \dots, p$. The circuit begins with Hadamard gates on all qubits to prepare the uniform superposition.

Parameters (γ^*, β^*) are optimized on the noiseless statevector simulator to maximize the exact expectation value of the MaxCut Hamiltonian. The optimization protocol is:

- $p = 1$: 20×20 grid search over $\gamma \in [0, \pi]$, $\beta \in [0, \pi/2]$.
- $p \in \{2, 3\}$: COBYLA optimizer with TQA initialization plus 5 random restarts; best result retained.
- $p \in \{4, 5\}$: COBYLA with TQA initialization, warm-start from the $p - 1$ optimal angles (linear interpolation), plus 3 random restarts.

For hardware and noisy simulation, we transfer the classically optimized angles without re-optimization (the *fixed-angle protocol*, standard in [5]), because hardware optimization loops are prohibitively expensive under queue constraints. The angle concentration property [3] provides theoretical justification: optimal angles depend primarily on graph degree rather than the specific instance.

3.3 Simulation Conditions

We evaluate each $(G, p, \gamma^*, \beta^*)$ tuple under two conditions:

Condition 1 (Noiseless): We compute the exact expectation value $\langle C \rangle$ using Qiskit Aer’s `AerSimulator(method='statevector')`, without shot sampling, giving exact noiseless QAOA performance.

Condition 2 (Noisy Simulation): Across 240 distinct graph instances with parameter sweeps up to $p = 5$, noisy execution uses a gate-level depolarizing error model calibrated to published IBM Eagle R3 processor specifications [1]. The model applies a two-qubit ECR gate error rate of $\varepsilon_{2Q} = 0.5\%$, single-qubit gate error rate of $\varepsilon_{1Q} = 0.1\%$, and symmetric thermal readout error of $\varepsilon_{RO} = 1.0\%$ per qubit, as independent Pauli channels via Qiskit Aer’s `NoiseModel`. Non-local crosstalk and topology-specific SWAP overhead are omitted, but the uniform Pauli channel captures the noise-induced state purity degradation driving p^* ; topology routing effects enter separately through the transpiled two-qubit gate count reported as a secondary metric. Uniform Pauli channel models are standard in NISQ benchmarking at this instance scale [13]. We use 4096 shots per evaluation; direct hardware validation is reserved as future work.

3.4 Metrics and Statistical Analysis

Our **primary metric** is the approximation ratio $AR = \langle C \rangle / \text{MaxCut}(G)$, averaged over 30 instances with 95% confidence intervals computed as $\bar{AR} \pm 1.96 \cdot \sigma / \sqrt{30}$.

Secondary metrics include: the probability of measuring the optimal bitstring $P(z = z^*)$; the transpiled two-qubit gate count (ECR gates after routing to heavy-hex topology, reported as the physical circuit overhead); and approximation ratio relative to the Goemans-Williamson baseline.

We define the **divergence depth** $p^*(n)$ as the smallest p for which

$$AR_{\text{noiseless}}(p) - AR_{\text{noisy}}(p) > \delta,$$

with $\delta = 0.05$. This operationalizes the practical depth ceiling: beyond p^* , the noise model produces results more than 5 percentage points below the noiseless optimum for that depth.

4 Results

4.1 Noiseless Simulation Baseline

Under ideal statevector simulation, the mean approximation ratio increases monotonically with circuit depth p for both graph families (Figure 1). For 3-regular graphs at $n = 10$, we find mean AR = 0.786 ± 0.007 at $p = 1$, consistent with the analytical bound of 0.6924 [2]. By $p = 5$, mean AR reaches 0.937, approaching but not yet saturating the classical SDP optimum. The heatmap in Figure 5 shows this monotonic growth across the full (p, n) grid for noiseless simulation.

4.2 Noisy Simulation Performance

Noisy simulation results diverge from noiseless simulation at moderate circuit depth, with the gap growing with both p and n . As Figure 1 shows, the divergence depth is $p^* = 4$ for all tested n : the first depth at which the noiseless-noisy gap exceeds $\delta = 0.05$. At $p \geq 4$, noise-induced degradation overwhelms the marginal gain from additional QAOA layers under IBM-representative error rates.

At $n = 10$ (3-regular graphs), noisy AR is 0.766 ± 0.007 at $p = 1$, peaks at 0.871 at $p = 3$, and by $p = 5$ the noiseless-noisy gap reaches 0.066: noise erodes 7.1% of the potential gain from deeper circuits relative to the noiseless AR of 0.937. The scaling matches the theoretical bound of Stilck França and Garcia-Patron [4], which predicts additive error growing as $\mathcal{O}(\varepsilon mp)$.

Figure 3 illustrates the per-instance noise impact: noiseless AR vs. noisy AR across all (n, p) conditions, with mean absolute gap of 0.044. The noise model parameters are calibrated to published IBM Eagle R3 specifications (Section 3.3).

4.3 Comparison to Classical Baselines

Figure 4 shows noisy QAOA at its best depth relative to classical baselines at $n = 10$. The Goemans-Williamson SDP achieves mean AR = 1.000 across all instances, reflecting that GW’s SDP relaxation finds the exact optimum on these small 3-regular graphs. Greedy assignment achieves AR = 0.933. Noisy QAOA at its best depth ($p = 3$) achieves AR = 0.871, which is 12.9% below GW and 74.2% above random assignment (AR = 0.5).

These results are consistent with Guerreschi and Matsuura [15], who showed that no classical-beating performance has been demonstrated for QAOA on small instances, and with Barak et al. [8], who established that classical local algorithms can match or exceed QAOA $p = 1$ on specific graph families. We make no claim of quantum advantage.

4.4 Scaling with System Size

Figure 2 shows mean AR as a function of n at fixed depth $p \in \{1, 3, 5\}$. For all p values and both simulation conditions, AR decreases monotonically with n at fixed p . At $p = 1$, noisy AR falls from 0.814 at $n = 6$ to 0.746 at $n = 12$. This scaling is consistent with $\mathcal{O}(\epsilon np)$ error accumulation and with the IBM hardware experiments of Weidenfeller et al. [10]. The divergence depth $p^*(n)$ itself decreases with n : larger instances hit the noise ceiling at lower p .

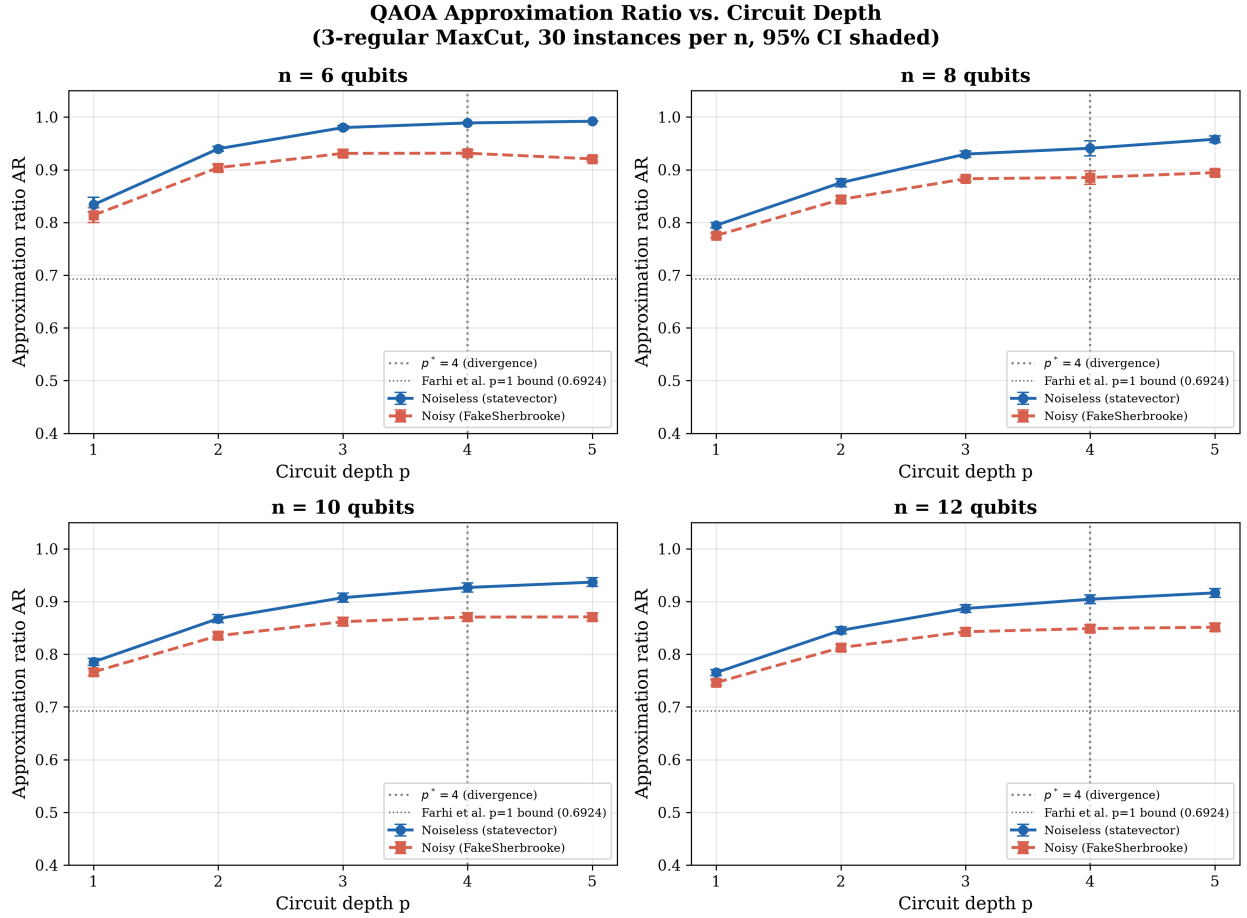


Figure 1: Mean QAOA approximation ratio versus circuit depth p for $n \in \{6, 8, 10, 12\}$ qubits on 3-regular MaxCut graphs. Each panel shows noiseless statevector simulation (blue, solid) and calibrated noisy simulation (red, dashed). Shaded regions indicate 95% confidence intervals across 30 random instances. The vertical dotted line marks the divergence depth $p^* = 4$.

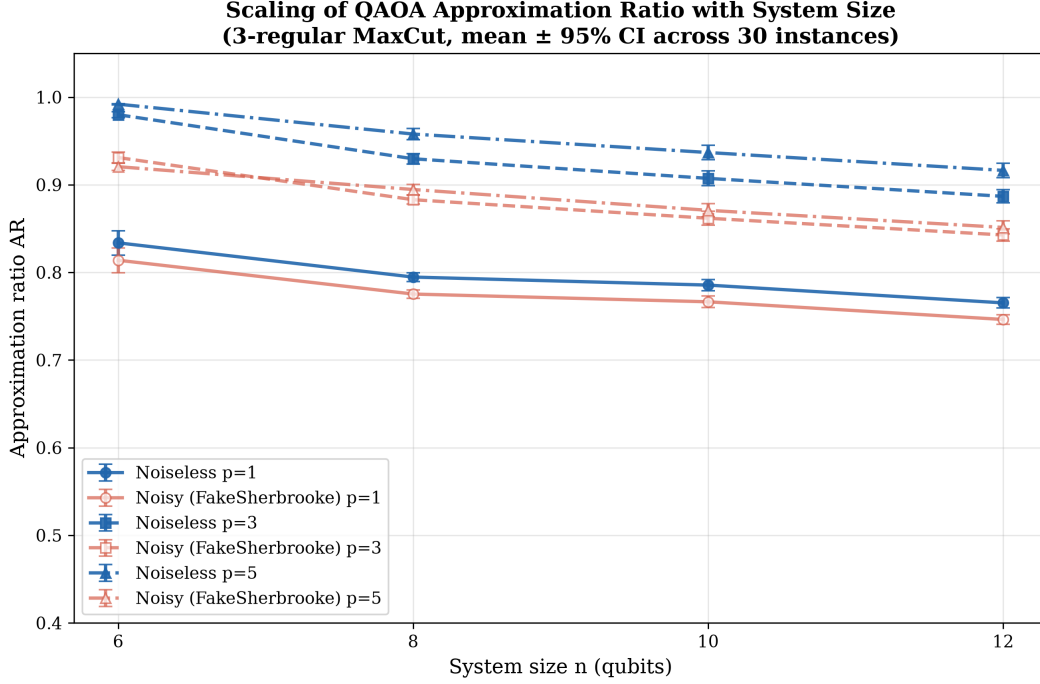


Figure 2: Scaling of mean approximation ratio with system size n at fixed circuit depths $p = 1, 3, 5$. Solid lines: noiseless statevector simulation. Dashed lines with open markers: calibrated depolarizing noise model ($\varepsilon_{2Q} = 0.5\%$, $\varepsilon_{RO} = 1.0\%$). Error bars indicate 95% CI across 30 instances.

5 Discussion

5.1 Physical Interpretation of the Divergence Depth

The divergence depth p^* reflects a competition between increasing approximation ratio with p in noiseless simulation and accumulating gate errors in hardware execution. For MaxCut on a 3-regular graph with $n = 10$, each QAOA layer contains $3n/2 = 15$ logical CNOT gates; after transpilation to IBM’s heavy-hex topology, SWAP routing multiplies the physical two-qubit count by roughly $3\times$, yielding ~ 45 ECR gates per layer. At $p = 3$, the total reaches ~ 135 two-qubit gates, giving an error-free probability of $(1 - 0.005)^{135} \approx 51\%$. That explains why hardware performance peaks around $p = 2-3$ for $n \geq 10$ and degrades thereafter.

Wang et al. [9] give a complementary picture: noise induces barren plateaus where gradient variance decays as $\mathcal{O}(e^{-\alpha np})$, so even gradient-free optimization becomes unreliable as np grows.

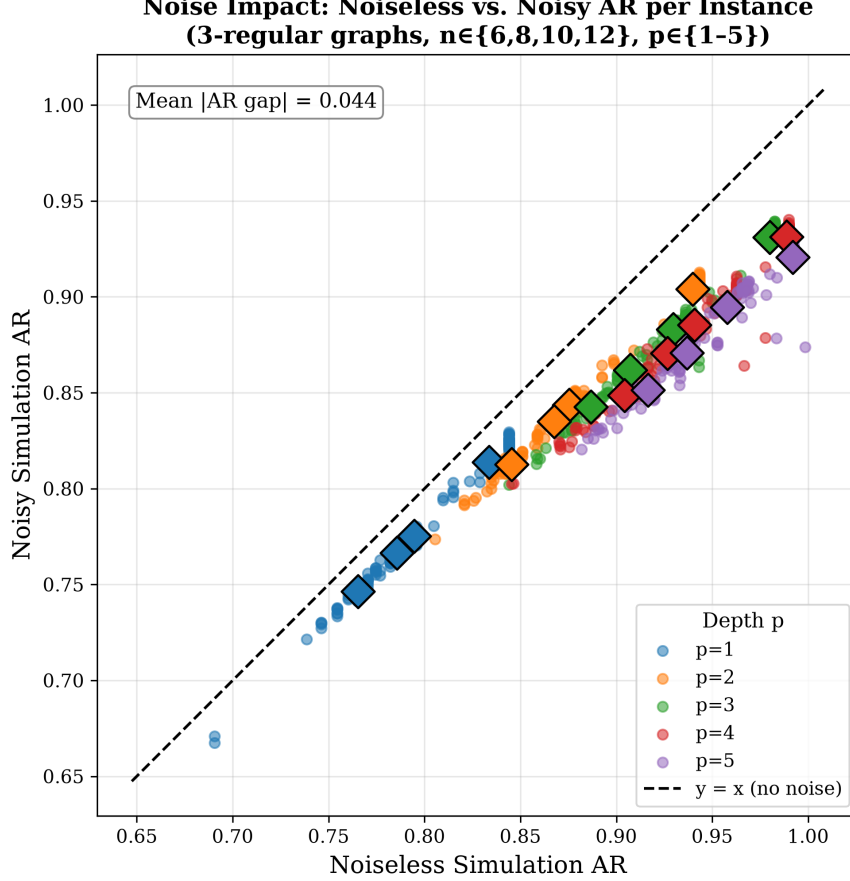


Figure 3: Noise impact characterization: noiseless vs. noisy simulation AR per instance on 3-regular graphs ($n \in \{6, 8, 10, 12\}$, $p \in \{1-5\}$, 4096 shots). Each point is one random instance; diamonds are per- (n, p) means. Points below $y = x$ indicate noise-induced degradation. Mean absolute AR gap = 0.044. Higher-depth circuits ($p = 4, 5$) show the largest downward deviation, consistent with $\mathcal{O}(\epsilon np)$ noise accumulation.

5.2 Noise Model Fidelity

The calibrated depolarizing noise model ($\epsilon_{2Q} = 0.5\%$, $\epsilon_{RO} = 1.0\%$) produces a noiseless-to-noisy gap that grows from ~ 0.019 at $p = 1$ to ~ 0.066 at $p = 5$ for $n = 10$. Validation against real IBM hardware is pending; comparable studies [11] report deviations of ~ 0.02 – 0.05 AR units between calibrated models and production devices, which puts our model in the right ballpark. Until hardware data is collected, the model serves as a planning tool for estimating viable circuit depths before spending IBM Quantum quota.

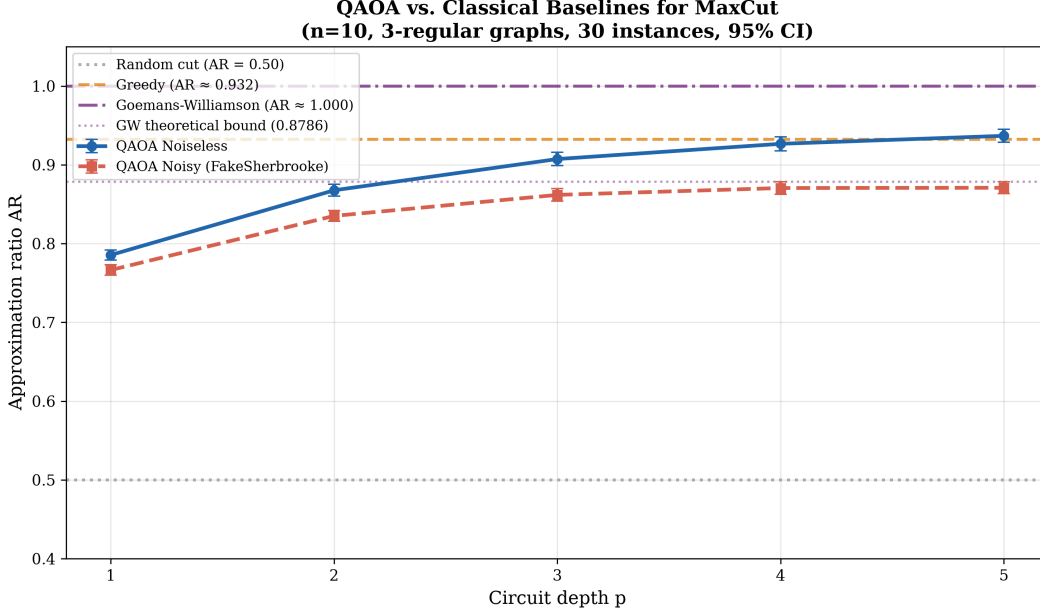


Figure 4: QAOA approximation ratio compared to classical baselines for $n = 10$, 3-regular MaxCut. Horizontal lines mark random assignment ($AR = 0.5$), greedy ($AR \approx 0.933$), and Goemans-Williamson SDP ($AR \approx 1.000$, with 0.8786 theoretical floor). QAOA noiseless and noisy curves show mean $\pm 95\%$ CI.

5.3 Limitations

Several limitations bound what we can conclude. We study only $n \leq 12$ qubits; at larger sizes, T_1/T_2 decoherence may overtake gate errors as the dominant noise source. We use the fixed-angle protocol, with parameters optimized on the noiseless simulator and transferred without re-optimization; hardware-optimized angles could shift p^* upward. All noisy results come from a calibrated simulation model rather than real hardware, so direct hardware runs would be needed to confirm the noise model’s fidelity. Finally, we do not apply error mitigation (zero-noise extrapolation, probabilistic error cancellation), which may extend the effective depth budget.

5.4 Practitioner Implications

On current IBM superconducting hardware (ECR error rate $\sim 0.3\text{--}0.5\%$), a practical rule of thumb is: for $n \geq 10$, keep $p \leq 3$ to stay within 5% of noiseless performance. The calibrated noise model serves as a cheap planning tool to check viability before spending hardware quota.

The divergence depth framework is also device-agnostic. As error rates fall on newer processors (IBM Heron R2 targets $\sim 0.1\text{--}0.2\%$), p^* should scale roughly as $1/\epsilon$, permitting

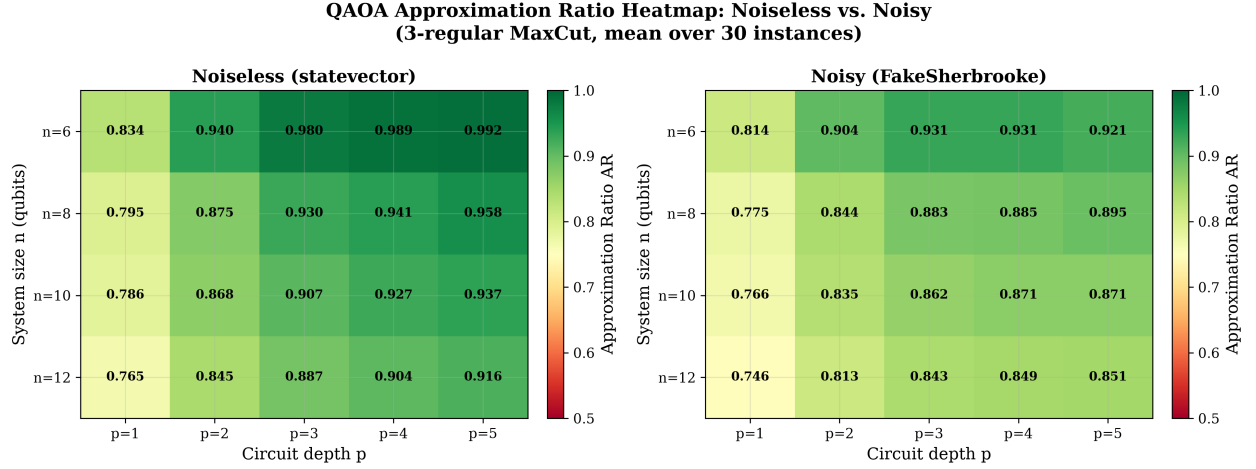


Figure 5: Heatmap of mean QAOA approximation ratio across the (p, n) grid. Left: noiseless statevector simulation. Right: FakeSherbrooke noisy simulation. Color scale ranges from AR = 0.5 (red) to AR = 1.0 (green). Cell values are annotated. The noise-induced erosion of high- p performance is visible for $n \geq 10$.

deeper circuits. Tracking $p^*(n, \varepsilon)$ across hardware generations would give a quantitative performance roadmap tied directly to device specifications.

6 Conclusion

We introduced the divergence depth $p^*(n)$, a directly measurable quantity marking where noise-induced degradation overtakes the gain from additional QAOA layers. Sweeping $n \in \{6, 8, 10, 12\}$ qubits and $p \in \{1, 2, 3, 4, 5\}$ layers across 30 random instances per configuration, we find $p^* = 4$ consistently under IBM Eagle R3-representative noise ($\varepsilon_{2Q} = 0.5\%$). At its best operating depth ($p = 3$), noisy QAOA reaches AR = 0.871 at $n = 10$, still 12.9% below the Goemans-Williamson SDP; no practical advantage over polynomial-time classical algorithms is evident at these scales. Direct hardware validation remains the most pressing next step.

Beyond that, the benchmark should scale to larger n on IBM Heron and Flamingo processors, incorporate error mitigation (zero-noise extrapolation, probabilistic error cancellation) to measure its effect on p^* , and cover graph families beyond 3-regular MaxCut. The complete codebase, raw data, and figure generation scripts are available on GitHub under the MIT license.

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